STAFF REVIEW · H446 133 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Use asymmetric establishment/authentication then symmetric traffic keys.

2. Sum 0, carry 1.

3. O(log n) for sorted random-access data.

4. On a match or after examining all items.

I Do — source worked steps

1. Trusted published digest — Find the given data and name the required result.

2. Calculate downloaded hash — Explain why this intermediate stage is necessary.

3. Compare exact digests — Calculate SHA-256 digest of received file and compare with the digest from a trusted source. If different, file changed or was corrupted. If equal, this strongly supports matching content, assuming trusted digest and secure algorithm; it does not by itself prove publisher identity.

We Do — guided CHECK

1. Choose random salt
Find the given data and name the required result.
Derive slow password hash
Explain why this intermediate stage is necessary.
Store salt plus result
Store unique salts and outputs from a slow password-hashing scheme such as Argon2id, not reversible encryption or a plain fast hash. On login, recompute using stored salt and compare securely.

You Do — source mark schemes

1. File integrity checking [2], password verification [2], hash table key indexing/fast lookup [2].

2. Distinct inputs can share a hash [2], finite output range implies possible collisions [2], robust use selects collision-resistant function/handles table collisions [2].

3. Hashing is one way not decryption [2], passwords use salting and suitable password KDF [2], digest comparison aids integrity [2], origin needs trusted channel/signature [2].

Exit Ticket — source answer

1. False. Hashing is not reversible encryption.